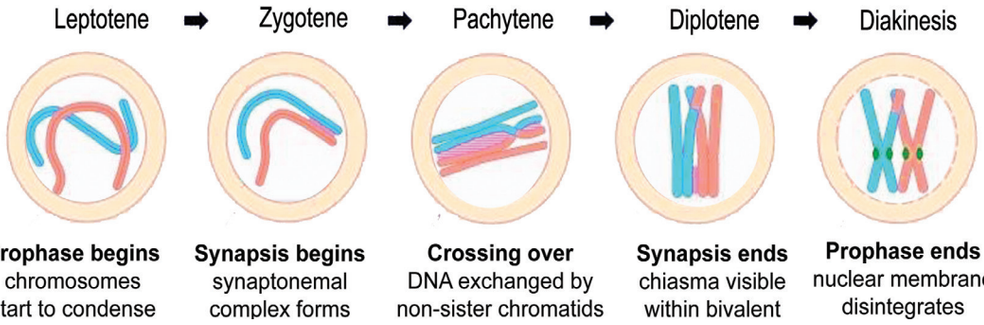



**Figure 7.7:** Prophase I

**Leptotene** – Chromosomes are visible under light microscope. Condensation of chromosomes takes place. Paired sister chromatids begin to condense.

**Zygotene** – Pairing of homologous chromosomes takes place and it is known as **synapsis**. Chromosome synapsis is made by the formation of synaptonemal complex. The complex formed by the homologous chromosomes are called as **bivalent (tetrads)**.

**Pachytene** – At this stage bivalent chromosomes are clearly visible as tetrads. Bivalent of meiosis I consists of 4 chromatids and 2 centromeres. Synapsis is completed and recombination nodules appear at a site where crossing over takes place between non-sister chromatids of homologous chromosome. Recombination of homologous chromosomes is completed by the end of the stage but the chromosomes are linked at the sites of crossing over. This is mediated by the enzyme recombinase.

**Diplotene** – Synaptonemal complex disassembled and dissolves. The homologous chromosomes remain attached at one or more points where crossing over has taken place. These points of attachment where 'X' shaped structures occur at the sites of crossing over is called **Chiasmata**. Chiasmata are chromatin structures at sites where recombination has been taken place. They are specialised chromosomal structures that hold the homologous chromosomes together. Sister chromatids remain closely associated whereas the homologous chromosomes tend to separate from each other but are held together by chiasmata. This substage may last for days or years depending on the sex and organism.

**Diakinesis** – Terminalisation of chiasmata, homologous chromosomes become short and condensed. Nucleolus and nuclear envelope disappears. Spindle fibres assemble.

**Metaphase I**

Spindle fibres are attached to the centromeres of the two homologous chromosomes. Bivalent (pairs of homologous chromosomes) aligned at the equator of the cell known as **metaphase plate**.

The random distribution of homologous chromosomes in a cell in Metaphase I is called **independent assortment**.

**Anaphase I**

Homologous chromosomes are separated from each other by shortening of spindle fibers. Each homologous chromosomes with its two chromatids and undivided centromere move towards the opposite poles of the cells. The actual reduction in the number of chromosomes takes place at this stage. Homologous chromosomes which move to the opposite poles are either paternal or maternal in origin. Sister chromatids remain attached with their centromeres.

**Telophase I**

Haploid set of chromosomes are present at each pole. The formation of two daughter cells, each with haploid number of chromosomes takes place. Nuclei reassembled. Nuclear envelope forms around the chromosome



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